

School of Engineering and Technology

K.R. Mangalam University
B.Tech CSE (AI & ML)
ENCS304 - Computer Networks
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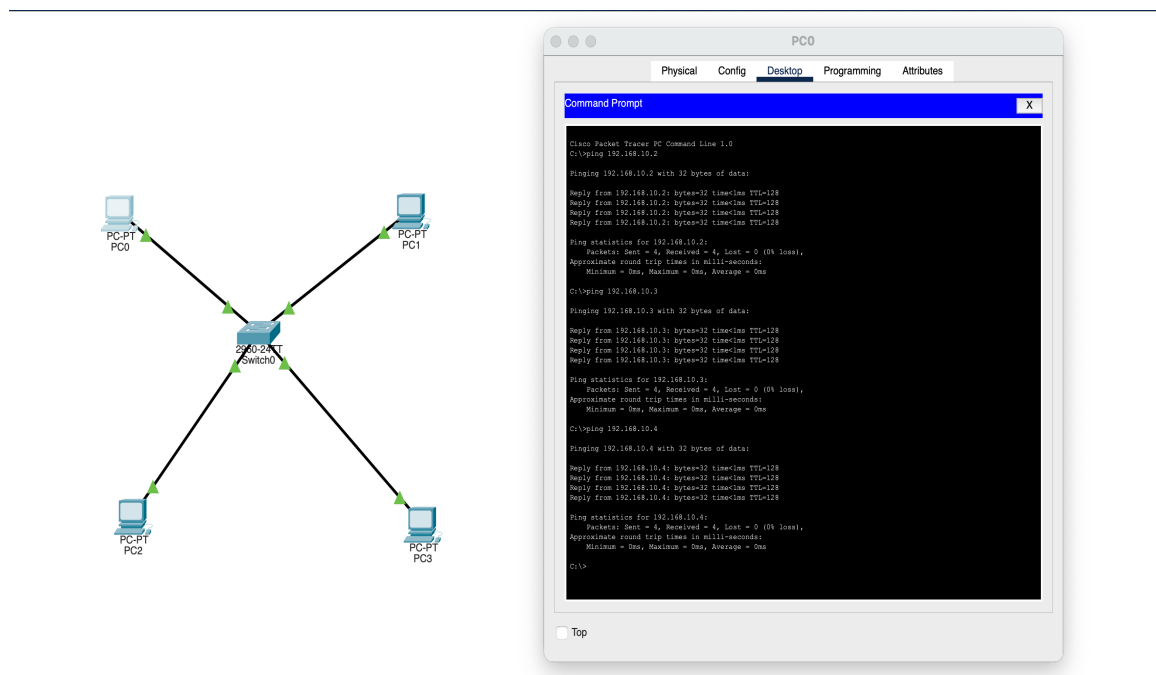
Assignment 1 Report - Basic Network Topologies

Objective:

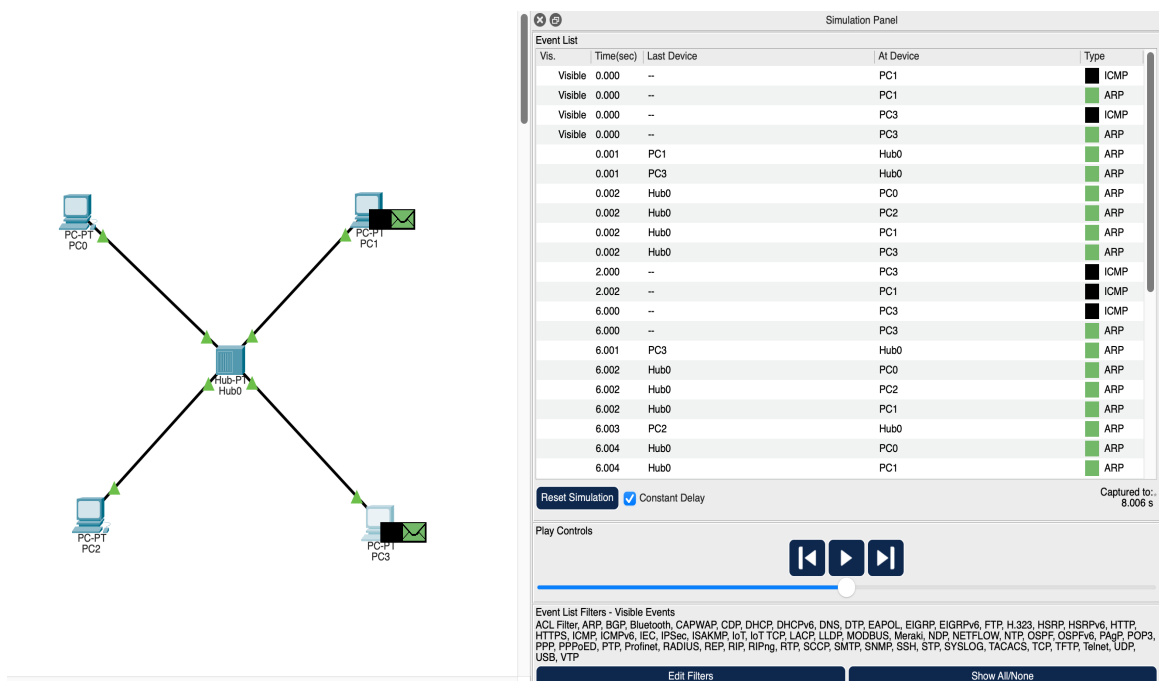
To design and analyze different LAN topologies using Cisco Packet Tracer and compare their connectivity, efficiency, and fault tolerance.

Topology Implementations:

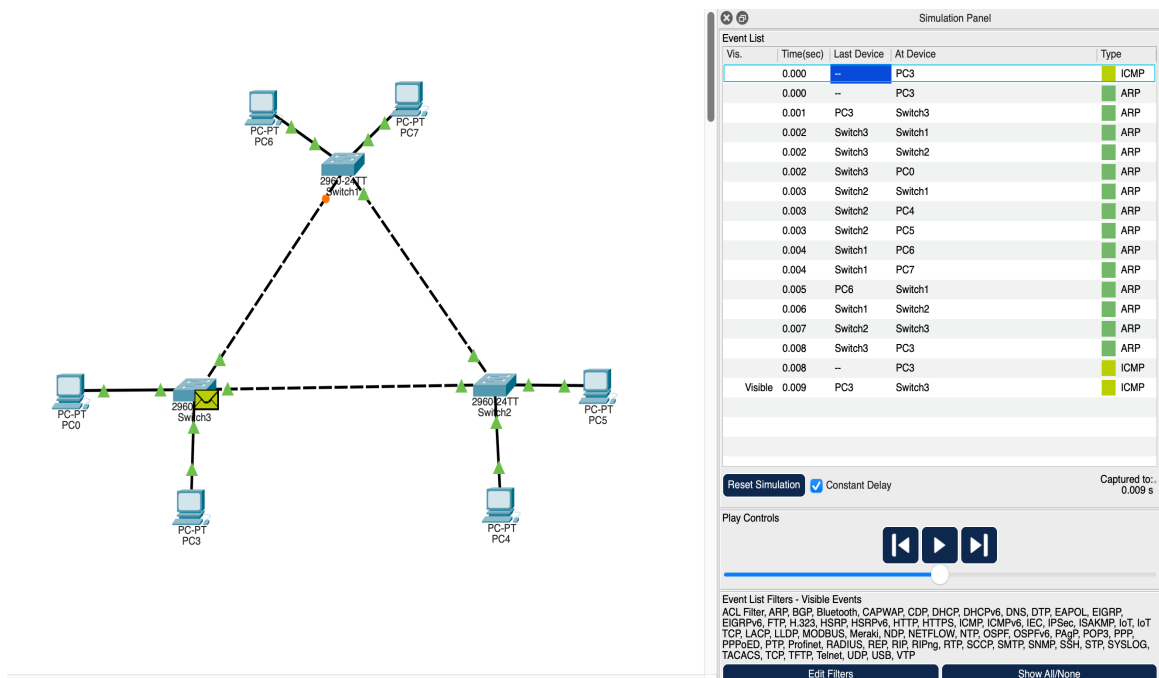
Task 1: Star Topology (Switch)



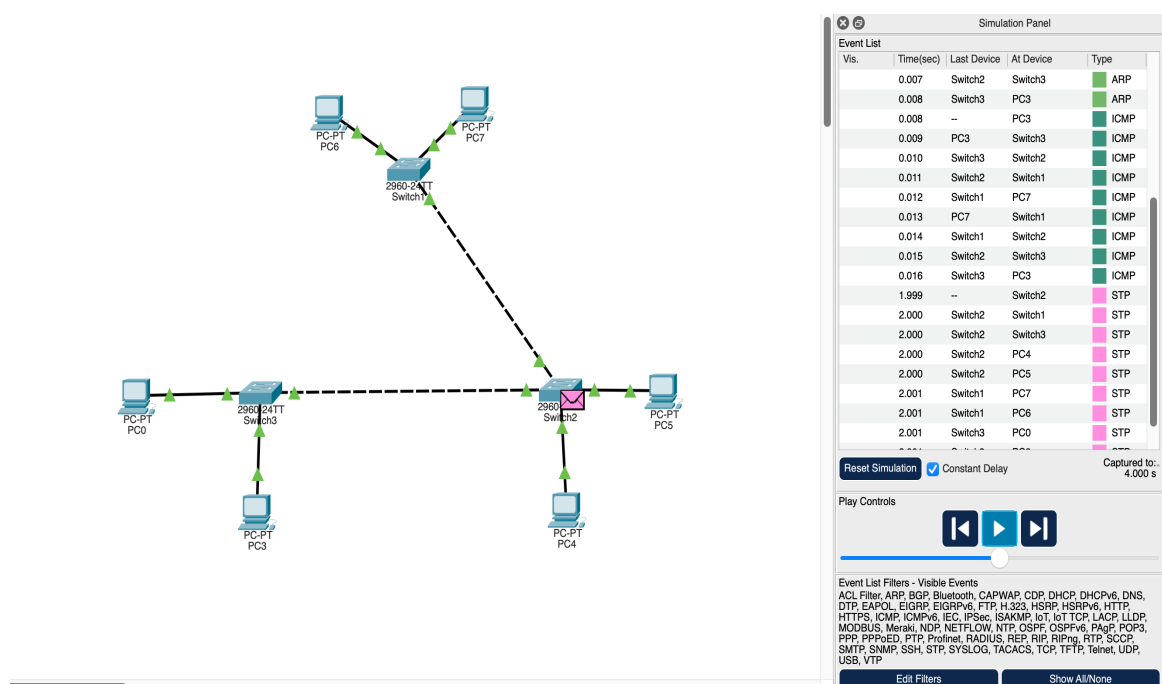
Task 2: Hub Topology



Task 3: Ring-like Topology



Task 4: Failure Test



Topology Comparison:

Topology	Device Used	Traffic Behavior	Fault Tolerance	Efficiency
Star	Switch	Forwards to specific MAC	Low (single point failure)	High
Bus-like	Hub	Broadcasts to all ports	Low	Low
Ring-like	3 Switches	Alternate path available	High (with redundancy)	Medium-High

Conclusion:

The switch-based star topology provides efficient communication with minimal unnecessary traffic. Hub-based topology causes broadcast traffic and is less efficient. The ring-like topology offers better fault tolerance due to alternate path availability. Selecting an appropriate topology is crucial for performance and reliability in LAN design.